How Microsoft Is Empowering Developers

David Ortinau

Principal Program Manager, .NET

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David Ortinau

@davidortinau

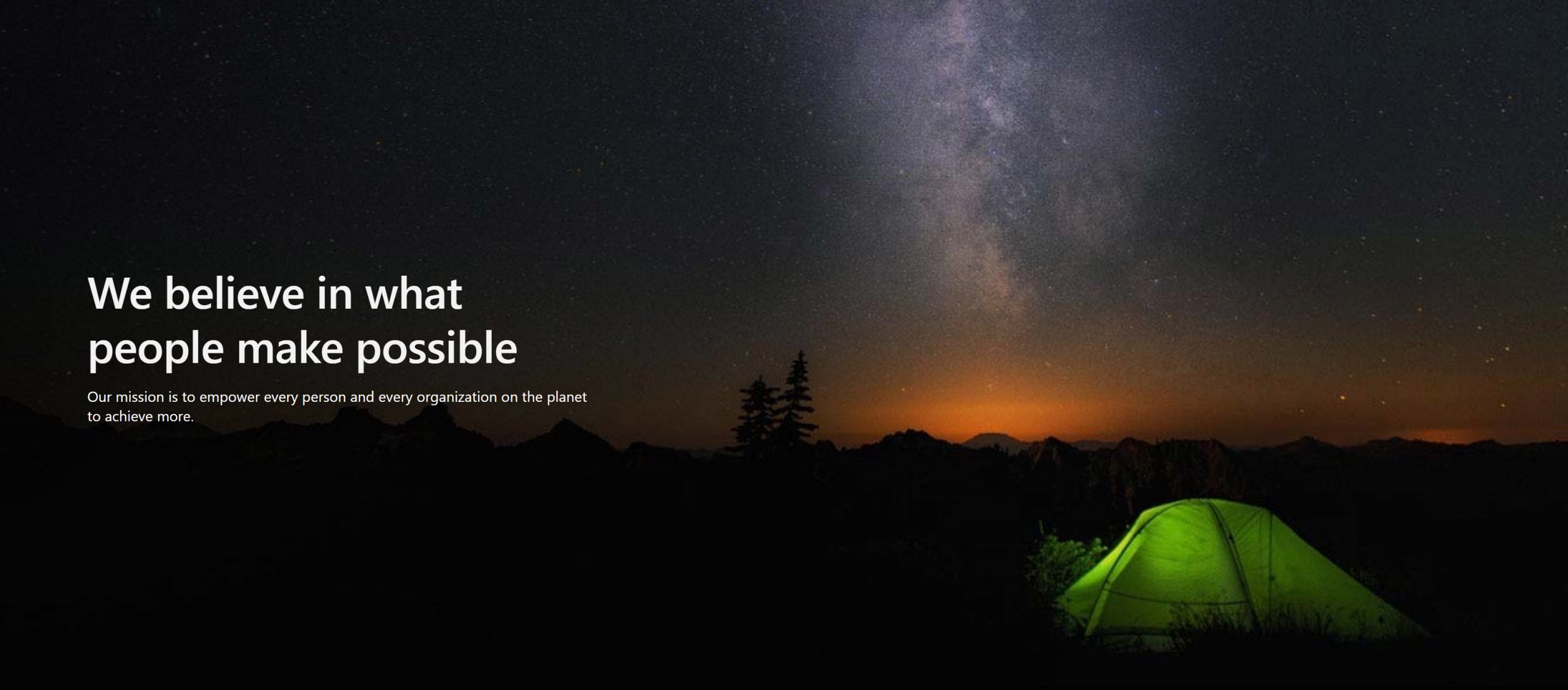
24 yrs. web, interactive, mobile

Xamarin MVP 2013 - 2017

BA English, Maryville University

Principal Program Manager
.NET MAUI

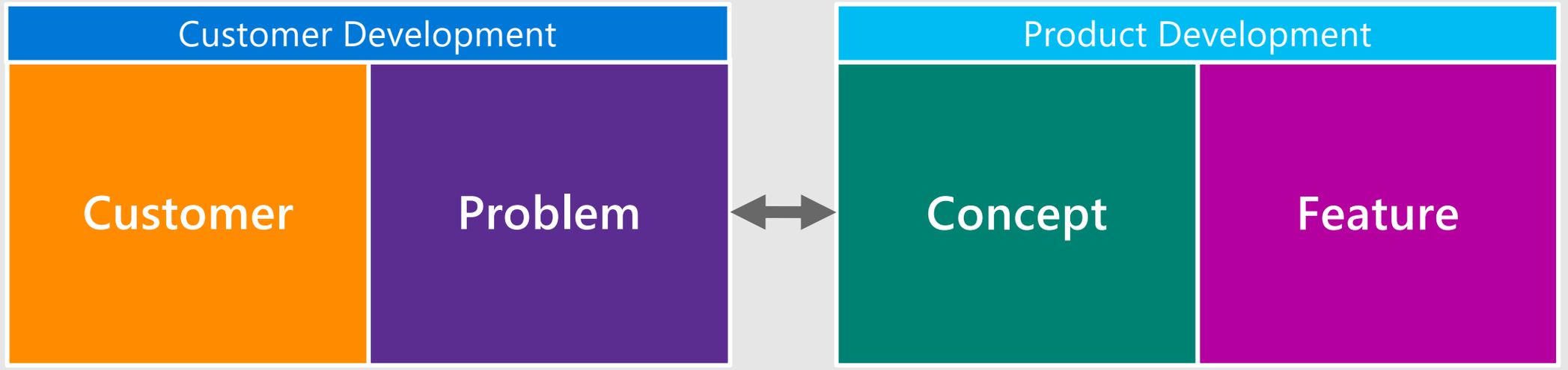
Microsoft



We believe in what people make possible

Our mission is to empower every person and every organization on the planet to achieve more.

Customer Driven Iteration



"What should we build?"

"How should we build it?"

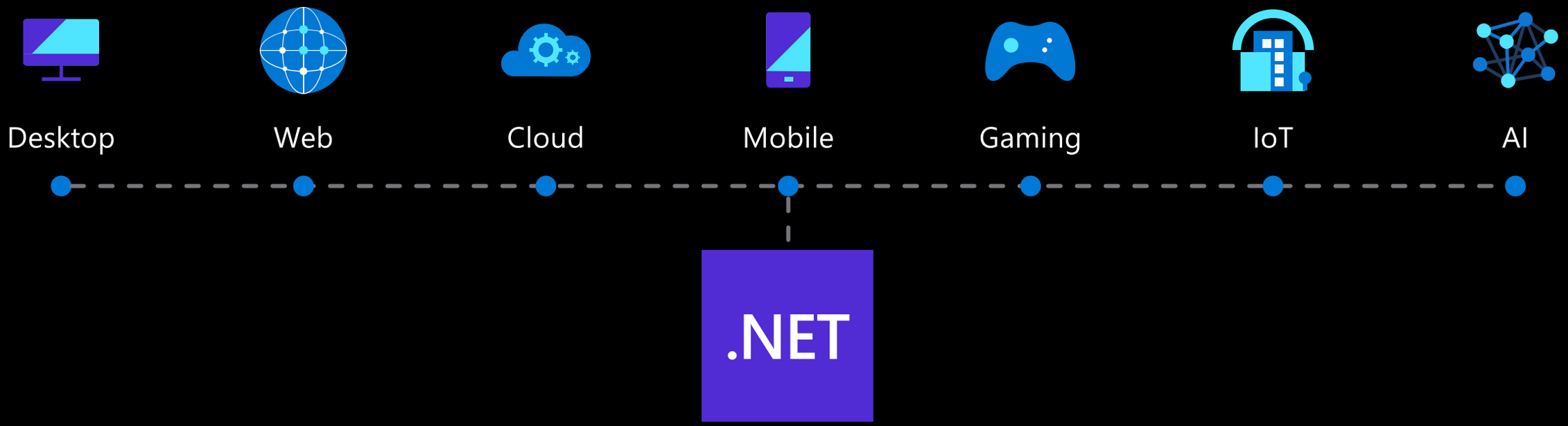
We work closely with customers every day, from 1:1 interviews to lab studies to large hands-on experiments to make sure we are constantly learning in order to improve products for you.

As such, we iterate quickly shipping as often as every 2 weeks.



.NET

Your platform for building anything



.NET – A unified development platform



CLOUD

Azure



WEB

ASP.NET

Blazor



DESKTOP

.NET MAUI

WPF

WinForms



MOBILE

.NET MAUI

Xamarin



GAMING

Unity



IoT

ARM32

ARM64



AI

ML.NET

.NET for Apache Spark

.NET 6

COMMON BASE LIBARIES/APIs

INFRASTRUCTURE

RUNTIME COMPONENTS | COMPILERS | LANGUAGES

ECOSYSTEM



NuGet



GitHub



Components, tools, library vendors

TOOLS



Visual Studio



Visual Studio for Mac



Visual Studio Code



CLI

You can write code with many .NET languages

C# (c-sharp)

- C# is a simple, modern, object-oriented, and type-safe programming language
- Its roots in the C family of languages makes C# immediately familiar to C, C++, Java, and JavaScript programmers

```
var names = new List<String>
{
    "Ana",
    "Felipe",
    "Emillia"
};

foreach (var name in names)
{
    Console.WriteLine($"Hello {name}");
}
```

F# (f-sharp)

- F# is a cross-platform, open-source, functional programming language for .NET
- It also includes object-oriented and imperative programming

```
let numbers = [ 1; 2; 3; 4; 5; 6; 7; 8; 9; 10 ]

let square x = x * x
let isOdd x = x % 2 <> 0

let squaresOfOdds =
    numbers
    |> List.filter isOdd
    |> List.map square

printfn "%A" squaresOfOdds
```

Visual Basic

- Visual Basic is an approachable language with a simple syntax for building type-safe, object-oriented apps

```
Dim names As New List(Of String)({
    "Ana",
    "Felipe",
    "Emillia"
})

For Each name In names
    Console.WriteLine($"Hello {name}")
Next
```

.NET ecosystem momentum

5.3M

Monthly active .NET developers

In Visual Studio family

#1

Most Loved Framework (2019, 2020, 2021)

.NET Core

Top 30

Highest velocity OSS project

github.com/dotnet

Top 5

Language on GitHub

C#

>10X

Faster than Node.js

ASP.NET Core

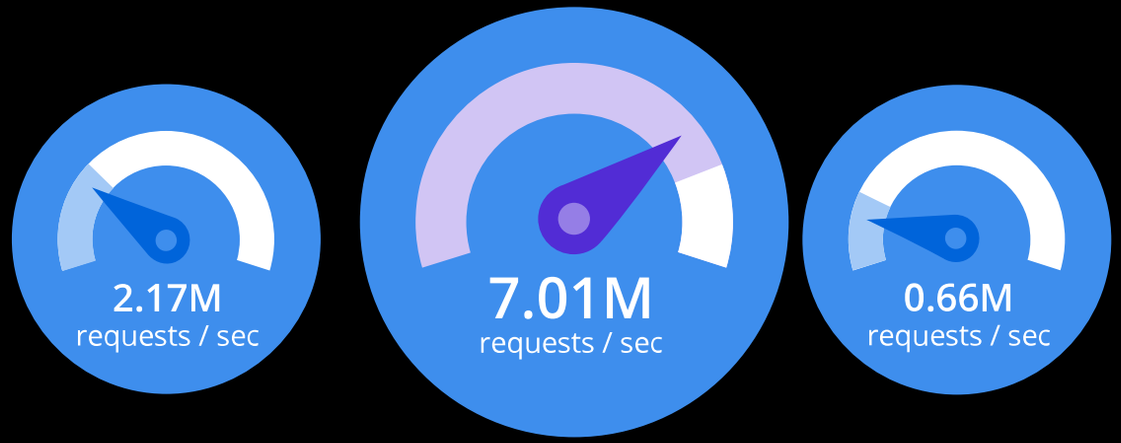
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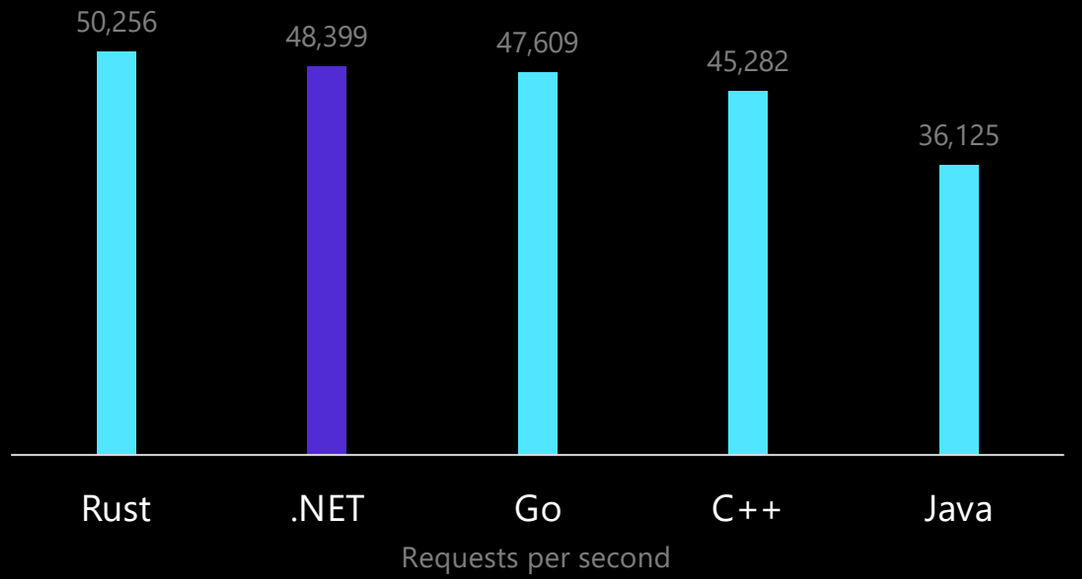
New to .NET are students

Download survey

Sources: insights.stackoverflow.com/survey/2019#most-loved-dreaded-and-wanted; insights.stackoverflow.com/survey/2020#most-loved-dreaded-and-wanted; github.com/cncf/velocity/blob/master/reports/top_projects_by_activity.txt; octoverse.github.com/; www.techempower.com/benchmarks/#section=data-r20&hw=ph&test=plaintext

.NET 5 Performance





>10X faster than Node.js

ASP.NET Core web framework

Faster than Go, C++ and Java

gRPC server performance

Sources:
https://www.techempower.com/benchmarks/#section=data-r20&hw=ph&test=plaintext
https://github.com/LesnyRumcajs/grpc_bench/wiki/2020-08-11-bench-results

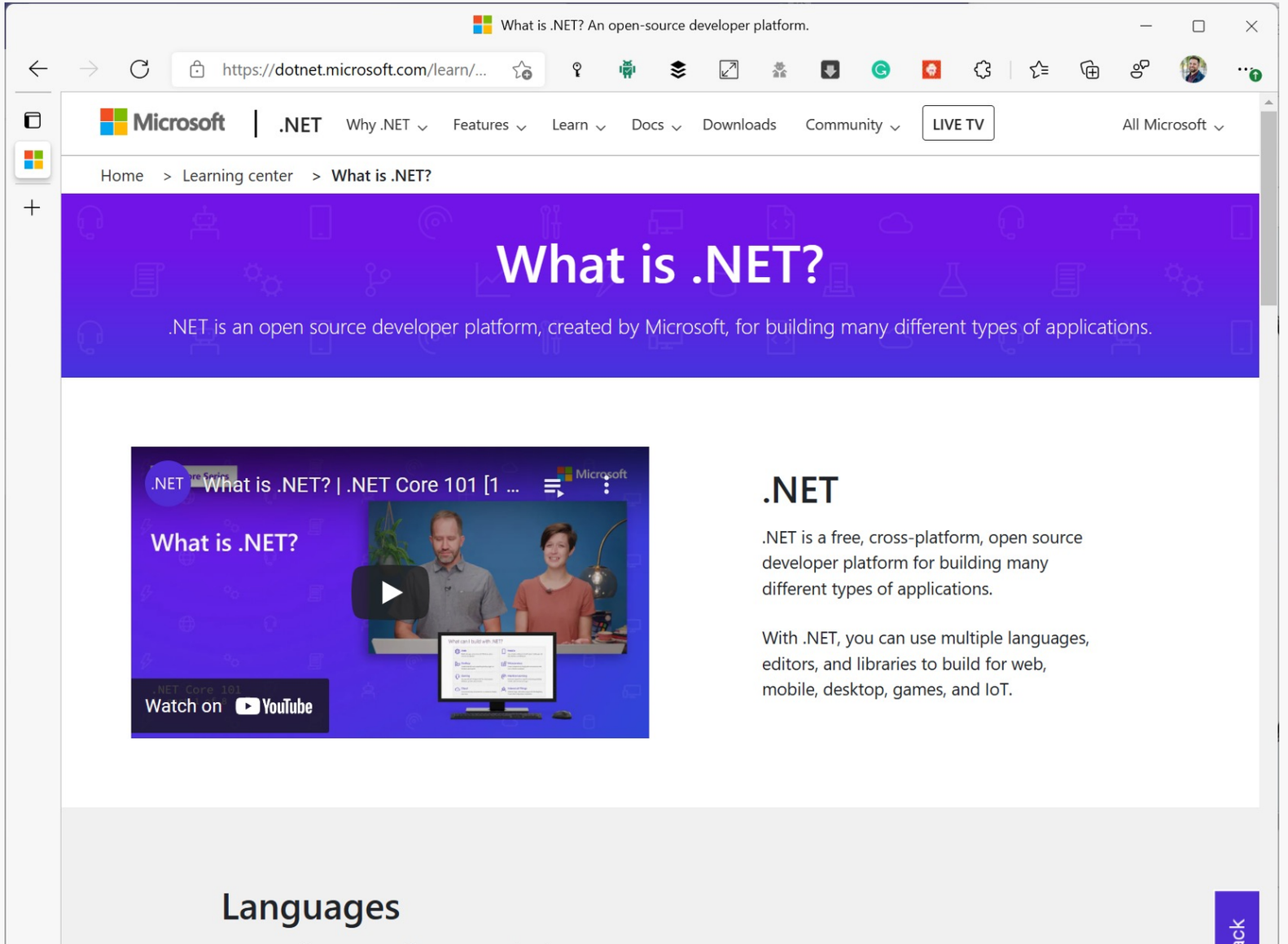
.NET

Loved by developers, trusted by enterprise

http://dot.net/customers



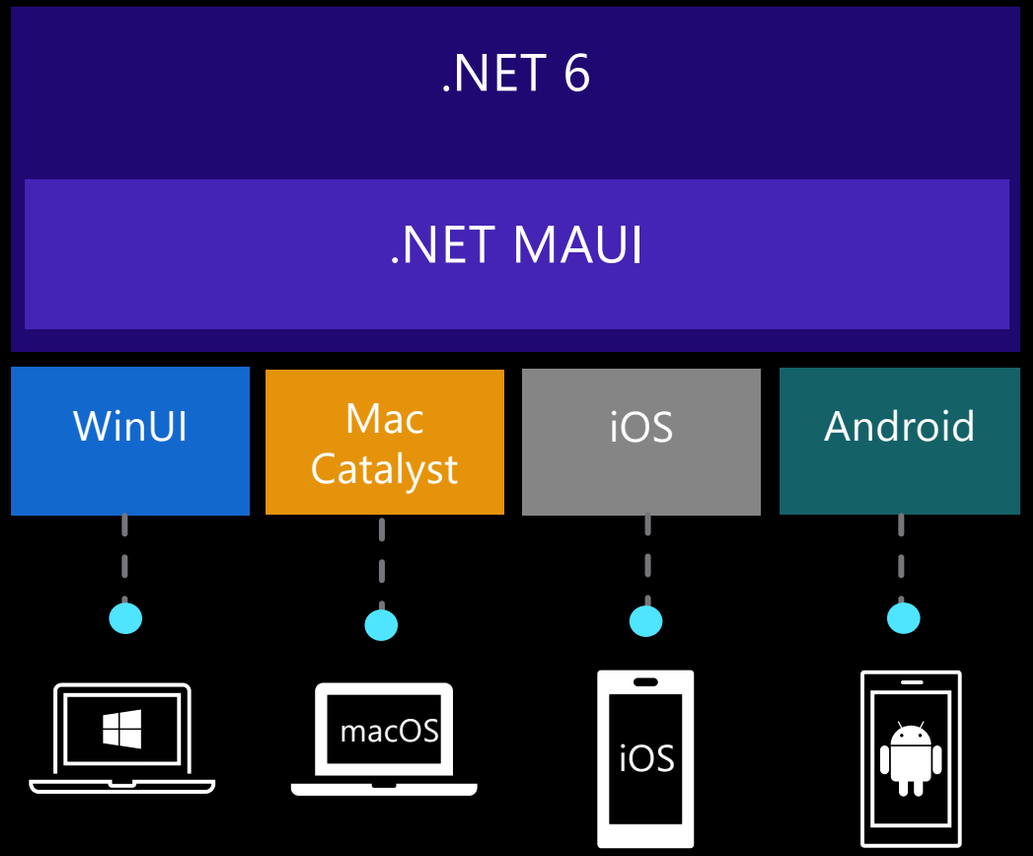
Learn more at https://dot.net



.NET MAUI is the most productive way to develop native apps that perform great on any device that runs Android, iOS, macOS, or Windows from a single codebase.

.NET Multi-platform App UI

Build native, cross-platform desktop and mobile apps with a single codebase



Cross-platform, native UI

Single project system, single codebase

Deploy to multiple devices, mobile & desktop

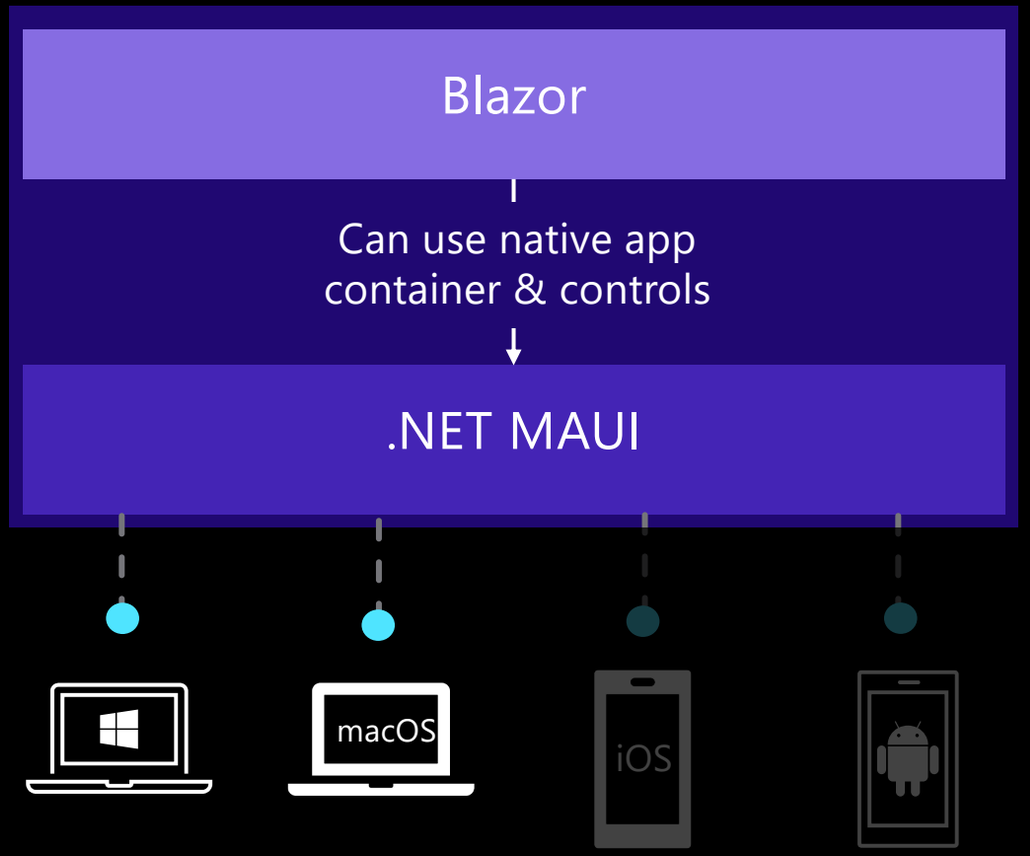
Evolution of Xamarin, upgrade path for users

General availability in Q2 2022

github.com/dotnet/maui

Blazor hybrid apps

Build high-performance, cross-platform desktop apps with C#, HTML and CSS



Reuse UI components across native and web

Built on top of .NET Multi-platform App UI

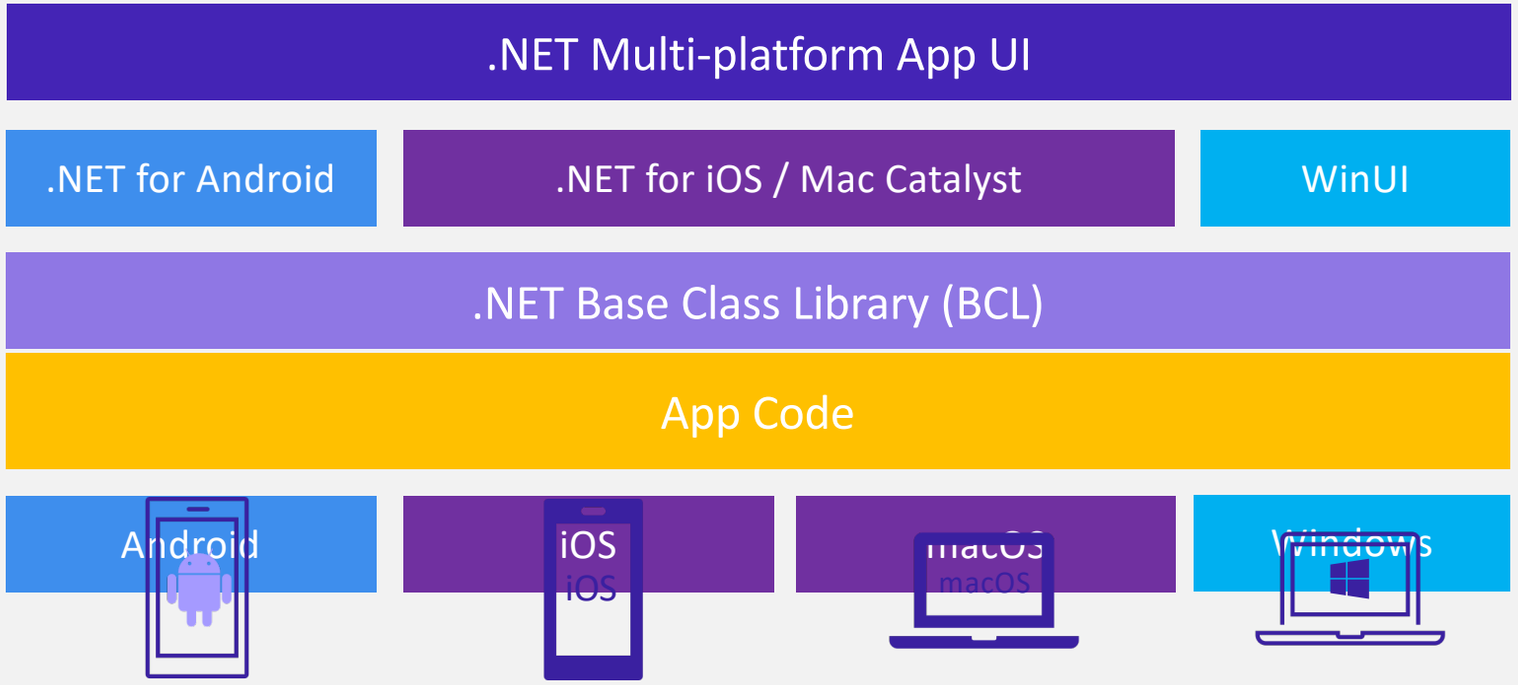
Native app container & embedded controls

Call native APIs, access file system, multi-threaded

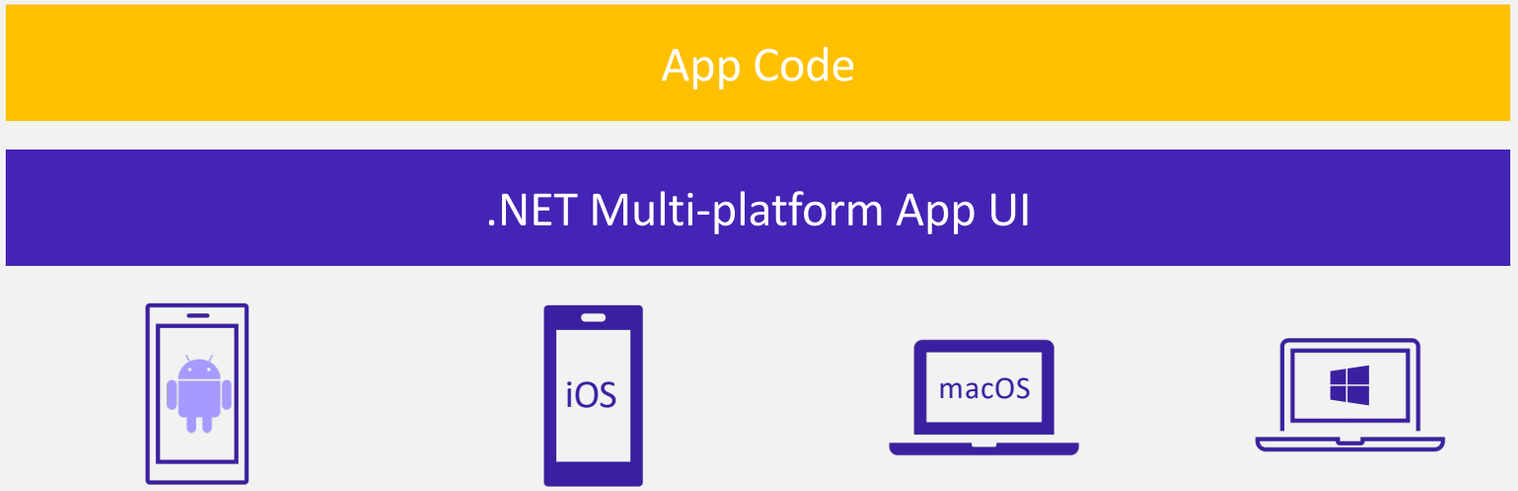
High-performance, optimal memory consumption


Weather TwentyOne
Wind Map
78%
From
Pond Elementary
0.2in
From
Pond Elementary
iPhone 12
iOS 14.5
3:58
Redmond, WA
Favorites
18°
Redmond, WA
USA
28°
St. Louis, MO
USA
WeatherTwentyOne
Redmond, WA
52°F
Clear
Winds
14|25
78%
From
Pond Elementary
0.2in
From
Pond Elementary
2%
From
Pond Elementary
9mph
From
Pond Elementary
78%
From
City Hall
0.2in
From
Rockwood
Reservation
2%
From
County Library
Next 24 Hours
Daily Forecasts
5:19
Redmond, WA
52°F
Clear
Next 24 Hours
Daily Forecasts

How does .NET MAUI work?



How does .NET MAUI work?



Do we have time?
Let's Demo!

https://github.com/davidortinau/WeatherTwentyOne



ANNOUNCING

.NET Conf 2021

November 9 – 11

.NET 6 launch

Free, three-day, virtual event

Save the date!

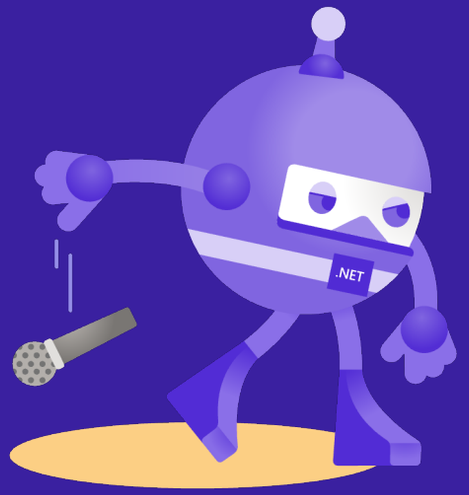
www.dotnetconf.net



Thanks for joining!

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Microsoft

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